

$$\begin{aligned}
 &= \{ (x,y) \mid (x \in A \wedge y \in B), \neg (x \in A \wedge x \in C) \text{ is True} \} \text{ (Defn. of negation)} \\
 &= \{ (x,y) \mid (x \in A \wedge y \in B), x \notin A \wedge x \in C \text{ is True} \} \text{ (Defn. of not belongs to)} \\
 &= \{ (x,y) \mid (x \in A \wedge y \in B), x \notin A \wedge x \in C \} \text{ (Defn. 8 from Set-2.1)} \\
 &= \{ (x,y) \mid x \in A \wedge y \in B \wedge x \notin A \wedge x \in C \} \text{ (Defn. of AND)} \\
 &= \boxed{A \times B - (A \times C)} \quad \square
 \end{aligned}$$

$$b) \overline{A \times (B \cup C)} = A \times (B \cup C)$$

Solution:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \quad A = \{1, 2, 3\} \quad B = \{4, 5, 6\} \quad C = \{7, 8, 9, 10\}$$

$$A \times (B \cup C) = \{(1, 5), (1, 6), (1, 7), (1, 8), (1, 9), (1, 10), (2, 5), (2, 6), (2, 7), (2, 8), (2, 9), (2, 10), (3, 5), (3, 6), (3, 7), (3, 8), (3, 9), (3, 10)\}$$

$$\overline{A \times (B \cup C)} = \text{The rest of 79 ordered pairs of } U \times U.$$

$$\overline{A} = \{4, 5, 6, 7, 8, 9, 10\} \quad \overline{B \cup C} = \{1, 2, 3\}.$$

$$\overline{A} \times \overline{B \cup C} \text{ has } 7 \times 3 = 21 \text{ ordered pairs.}$$

$$\therefore \overline{A \times (B \cup C)} \neq \overline{A} \times \overline{B \cup C} \quad \square$$

The symmetric difference of A and B, denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B.

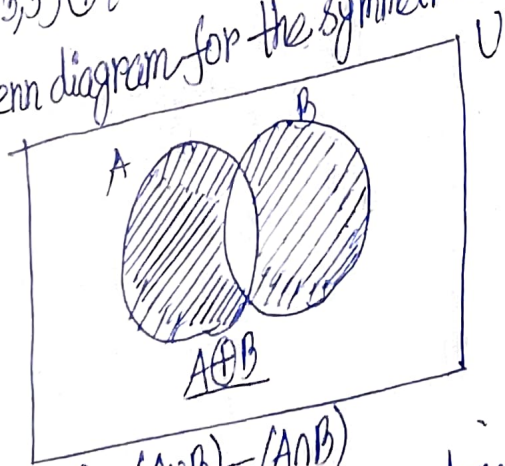
38) Find the symmetric difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$

$$\{1, 3, 5\} \oplus \{1, 2, 3\} = \{2, 5\} \text{ (By defn.)}$$

Solution:

40) Draw a Venn diagram for the symmetric difference of the sets A and B.

Solution:



41) Show that $A \oplus B = (A \cup B) - (A \cap B)$

Solution:

Let, x be an arbitrary element of $A \oplus B$, i.e. $x \in A \oplus B$.
By Defn. of membership, an element can either be a member of a set or not be a member of a set.